

William Chen

832-488-3816 • wchen23@nd.edu

Research Experience (PhD) – University of Notre Dame

2021 - Present

Wastewater-Based Epidemiology (WBE) and modeling:

- Independently designed and conducted WBE experiments on the malaria-causing agent in the US and Kenya
- Devised a model-based framework to assess the probability of detection using WBE on any disease (Monkeypox, Zika)
- Compiled data and conducted extensive research on pathogenic shedding, wastewater flow rates, and historically reported case rates in constructing the model-based framework for Monkeypox, Zika, Dengue, Malaria, Tuberculosis, etc.
- Conducted campus wastewater monitoring of SARS-CoV-2 – sampling, RNA extraction, and RNA ddPCR

Expected graduation: December 2024 - advanced from PhD student to candidate on April 10, 2023

Cumulative GPA – 4.0

Research Experience - Quantitative Microbial Risk Assessment (QMRA) IV

July - Aug 2022

- Led a group of 6 at the workshop for 1 week with modeling, coding, presenting, discussing, and writing for future QMRA-related research publication on the fungal Valley Fever disease

Fellowships, Awards, & Publications

2021 - Present

- Recipient of the University of Notre Dame's Select Dean's Fellowship
 - Recipient of the Borchardt Conference's 26th Triannual (2023) Certificate of Excellence Poster Presentation Award
1. **Chen, W.**, & Bibby, K. (2023). Making Waves: Establishing a Modeling Framework to Evaluate Novel Targets for Wastewater-Based Surveillance. *Water Research*, 120573.
 2. **Chen, W.**, & Bibby, K. (2023). A Model-Based Framework to Assess the Feasibility of Monitoring Zika Virus with Wastewater-Based Epidemiology. *ACS ES&T Water*, 3(4), 1071-1081.2.
 3. **Chen, W.**, & Bibby, K. (2022). Model-based theoretical evaluation of the feasibility of using wastewater-based epidemiology to monitor monkeypox. *Environmental Science & Technology Letters*, 9(9), 772-778.3.
 4. Crank, K., **Chen, W.**, Bivins, A., Lowry, S., & Bibby, K. (2022). Contribution of SARS-CoV-2 RNA shedding routes to RNA loads in wastewater. *Science of The Total Environment*, 806, 150376.

Submitted (Potential future publications):

1. Khan, D., **Chen, W.**, Linden, Y., Corbeil, K., Lowry, S., Higham, C., Mendez, K., Burch, P., DiFondi, T., Verhoughstraete, M., Galgiani, J., Haas, C. N., Gerba, C., Hamilton, K. A. (2023). A microbial risk assessor's guide to Valley Fever (*Coccidioides* spp.): Case study and review of risk factors. *Science of The Total Environment*.

Education (PhD): Engineering Research Courses – University of Notre Dame

2021 - 2022

Water, Disease, and Global Health:

- Presented and wrote a research article on a QMRA study about soilborne exposure pathways of Bangladesh children
- Learned about various global diseases including water-borne, vector-borne, air-borne, venereal, and zoonotic diseases

Bacterial Communities:

- Conducted a literature review of over 30 research papers on bacterial influence on agricultural yield
- Prepared weekly presentations and led discussions on topics relating to microbiology and microbial communities

Environmental Microbiology:

- Learned about microbial metabolism, division, gene expression/regulation, culturing methods, molecular analysis such as PCR and sequencing, and applications like epidemiology, QMRA, and wastewater monitoring
- Researched, presented, and wrote a report about *Giardia lamblia* as a group project

Geomicrobiology:

- Led 2 one-hour-long presentations about the microbial influence on geologic activities: weathering and acid-mine drainage

Education (Masters): Engineering Research Courses – Stanford University

2016 - 2018

Coastal Contaminants Course:

- Conducted weekly assignments involving compiling, deciphering, and comparing seasonal tide/wave elevation, water temperature, and biomass data from online databases into graphs, tables, and correlation formulas
- Independently utilized information from research papers and online databases to answer thesis on final project about the effects that water temperatures have on the development of algal blooms and red tides

Remote Sensing of the Oceans Course:

- Used satellite imaging programs QGIS and SeaDAS to further investigate the effects that ocean temperatures have on the growth of algal blooms in coastal areas, especially as it relates to the geospatial distribution of algal biomass

Environmental Planning Methods Course:

- Acted as the Environmental Protection Agency's manager in moderating negotiations with stakeholders in a simulation focused on addressing a Superfund problem related to river pollution
- Researched and presented the importance that environmental leadership has on corporate and consumer opinion in environmental stewardship and green marketing

Introduction to Human Exposure Analysis Course:

- Worked in a group of 3 in preparing for the final project about the effects that flushing toilets have on aerosols' ability to carry pathogens into the air and presented the team's finding with comprehensive graphs using data from many papers
- Practiced reviewing and critiquing other research papers and classmates' presentations and improved own researching strategies based on the feedback received from others

Education (Bachelor's): Engineering Design Courses – UCLA

2009 - 2014

Foundations & Earth Structures Design Course:

- Led a team of 5 to design foundations and levees to assess the risks and feasibility of constructing structures around a proposed area that possesses significant problems (an island with soil that has high water content)

Structural Mechanics Laboratory Design Course:

- Used computer programming to graph the stress vs. strain relationships and the damping effects for various experimental trials; explained possible reasons for discrepancies between trials by analyzing the graphs
- Researched and presented the damping and noise disturbance effects relating to structural/mechanical tests

Design Improvement Project: Engineering Ethics Course:

- Collaborated with a team of 6 to improve the design of a watch using knowledge from every engineering discipline
- Researched and wrote a paper about eutrophication's effect on natural habitat, ecology, and human health
- Researched and wrote a paper about the impact that ethics had on promoting or avoiding levee failures in 2 cases

Work Experience – University of Notre Dame

Teaching Assistant for Introduction to Environmental Engineering Laboratory Course:

- Assisted students in performing laboratory procedures for 6 laboratory experiments - prepared physical and chemical laboratory setups and helped troubleshoot when problems arise
- Graded and provided detailed feedback on laboratory reports, weekly homework, and exams for 20+ students
- Held weekly office hours helping students excel in coming up with unique solutions to homework problems

Work Experience – Stanford University

Teaching Assistant for Master's Level Course – Aquatic Chemistry:

- Led comprehensive weekly problem/discussion sessions that provided alternative strategies for understanding the material
- Created exam/course-book problems that challenged students to apply chemistry in real life such as determining the pH of blood and calculating the cadmium concentrations of rice patties and human blood after exposure to polluted water

Data Analyst/Surveyor/Research Assistant for Professor:

- Independently organized, assessed, and presented 10 years' worth of daily precipitation data using time-series analysis
- Set up the equipment, read/interpreted the measurements, and recorded the data for level-field surveying

Work Experience – United States Navy

Prototype and Classroom Instructor:

- In charge of teaching other naval officers how to operate and program simulated submarine prototypes
- Instructed mathematics and science material to other naval officers and graded homework and exams
- Trained military personnel on the operation procedure of nuclear power plants and how to troubleshoot technical errors
- Excelled in intensive training program covering the physics, mechanics, and operation of nuclear submarine powerplants
- Developed leadership skills as a naval officer in following through with disciplinary and moral codes, working in a team as a cohesive unit, exchanging knowledge with other officers, and inspiring future junior sailors

Work Experience – BCEGI/GHJ Construction Inc., LLC

Project Estimating and Engineering:

- Analyzed architectural, structural, and civil engineering drawings to determine the requirements and specifications for 3 major development projects
- Compiled multiple 90-page contracts, bid comparisons, change orders, RFIs, schedules, and pay applications to ensure that the project deadline is met
- Inspected the project site on a regular basis to ensure that problems do not slow down the progress of construction
- Updated and closely monitored construction schedules, coordinated with city engineers to obtain building permits, and communicated with multiple subcontractors to ensure proper construction procedures were met
- Conducted real estate and demographic-related research on twenty upcoming projects
- Organized and edited presentation slides for eleven projects sent to the company's headquarters

Volunteering Work Experience – Runa Foundation: Nonprofit Organization in Ecuador

Forest Restoration/ Environmental Conservation Intern:

- Led a group of 6 interns in navigating around a wide span of forest/plantation plots - in charge of implementing navigation tools such as GPS devices, drawing maps by hand, and interpreting ArcGIS maps in location finding
- Primarily responsible for managing the collection, organization, and analysis of mapping data in the ArcGIS program
- Independently redesigned various presentation slides and financial analysis spreadsheets to ensure that the information is understandable and attractive to both professionals and native community members

Skills

Computer Skills:

- R Studio – Monte Carlo simulation, fitdistr mle, ggplot graphing, statistical analysis, sensitivity analysis
- QuantaSoft – ddPCR reader (BioRad) running, threshold setting, and data extraction and analysis
- MATLAB, Python, AutoCad, ARCGIS, Microsoft Excel/Project/Word/PowerPoint, C++ programming, PlanSwift,

Laboratory Skills:

- DNA/RNA extraction – AllPrep PowerViral DNA/RNA Kit (Qiagen), bead-beating, centrifuging, vertexing
- Sampling of campus wastewater – programming autosampler & conducting weekly sample collection
- DNA/RNA ddPCR/dPCR – reaction preparation, droplet generation, thermocycling, droplet reader (BioRad/QIAcuity)

Certificates

- Mastery of AutoCAD – UCLA extension course and completion of Udemy class
- R Studio certificate of completion from Coursera, Data Camp, and Udemy online courses